

How do Users Expect their Smart Memory Vaults to Utilize Their Shared Memories?

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	Customizing ads	Cuing real interactions	Customizing news feed	Third-party sharing
REASON OF SURPRISE				
Personal data exploitation				
Lack of informed consent				
Inexplicable platform behavior				
Out-of-scope usage objective				
EFFECT OF THE INCIDENT				
Awareness of platform data usage pattern				
Discussing phenomenon with others				
Negative company image				
Reducing platform usage				

Figure 1: The columns show the most remembered Smart Interactions involving Personal Memories (SIPMs) on Facebook (FB) by Egyptian users. The rows show why the users are apprehensive about the interaction and how it impacted their perception of the platform. We use FB as an example for modern smart semi-public memory vaults.

ABSTRACT

Social media (SM) is a popular accessible form of smart memory vaults. Prior work shows that users are still unable to understand how their shared memories are used by smart systems to create Smart Interactions involving Personal Memories (SIPMs). This can lead to negative social repercussions such as cyberbullying. This

work investigates the most memorable SIPMs on Facebook for Egyptian users and their impact on platform usage. We conducted an online survey (N=53) requesting critical incident reports about surprising Facebook SIPMs. The most remembered SIPMs were: customizing advertisements, cuing offline interactions, sharing data with third parties and personalizing the newsfeed. Our results suggest that SIPMs, particularly customized advertisements act as ambient memory augmentation solutions to the users' shared memories. Additionally, negative platform perception does not necessarily translate into reduction of platform usage. Our work inspires the discussion about users expectations towards ambient usage of their data on smart memory vaults.

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CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519826>

CCS CONCEPTS

• **Human-centered computing** → **Social media; Ambient intelligence; Empirical studies in HCI.**

KEYWORDS

social media, critical incident report, affect, user experience, informed consent, privacy, advertisements, memory vaults, personalization

ACM Reference Format:

Passant Elagroudy, Mennatallah Saleh, Christian Sturm, and Albrecht Schmidt. 2022. How do Users Expect their Smart Memory Vaults to Utilize Their Shared Memories?. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3491101.3519826>

1 INTRODUCTION AND RELATED WORK

Social media is a modern accessible form of memory-altering systems (see [14] for an example application). Users share selective snippets of events and interests to curate a digital archive of their lives. It is estimated that 49% of the world population currently uses social media, which is equivalent to 4.15 billion mobile users world wide, with Facebook being the most commonly used platform [17]. The wide proliferation offers a unique opportunity for understanding users' perspective on privacy and preferences of sharing their memories with other users and/or smart platforms to fuel intelligent personalized interactions. Within this manuscript, we refer to these interactions as *SIPM*, standing for Smart Interactions involving Personal Memories. We consider any content explicitly shared or interacted with on social media as a cue to a memory and refer to it here as shared memory.

There are several theories about reasons for sharing on social media (see [11] for a concise review). In essence, talking about oneself is a rewarding task that helps in self representation [16]. However, negative repercussions of sharing such as cyber-bullying and radicalization (e.g. [12, 13]) are maximized when the users do not understand how their data is being used on the platform and/or who can see and interact with it. For example, users found accurate automated personality analysis from their social media interactions “creepy” and they felt helpless as they had the urge to share it but did not like this aspect [18]. Prior work has also found that the majority of users are unaware about the implications of their privacy settings [1, 9]. Similarly, Eslami et al.[3] found that over half of the users were unaware that the Facebook news feed is algorithmically curated. Hsu et al. [8] also found that half of their participants randomly set their privacy settings despite having a need to set it correctly because: (1) they either did not understand it or (2) were not aware it exists, particularly ones related to advertisement personalization. Therefore, further research is required to understand how users perceive SIPMs, as they may include sensitive information that exposes the user's virtual activities.

We extend the aforementioned body of work by conducting a preliminary online survey (N=53) to understand what users in non-WEIRD communities remember about SIPMs on Facebook (FB). We use critical incident recalling to analyze incidents, their reasons

and impact on platform usage in order to understand the user expectations about how smart memory vaults implicitly use their shared memories. Our contribution is primarily coming from the representation of our user group (Egyptian nationals) as literature focusing on privacy and behavior on social networks is underrepresented in non-WEIRD countries (e.g. [2, 4, 7]). We use FB as an example medium because it is widely used in our user group as 35 out of 50 million of the Egyptian internet users are active Facebook users [10, 15]. Additionally, it supports several media formats, thus rendering rich memory cues. Lastly, it is a common example of an AI¹-powered social media platform. Specifically, we look at the effect of the surprising SIPMs on the affect and the behavioral change resulting from using the platform. *Surprising incidents* within our context are defined as unexpected behaviors on the platform regardless of their connotation. This means that a user can be excited about a surprising incident or appalled by it.

Our results suggest that the most remembered SIPMs are: customized advertisements, cuing of real world interactions such as “people you may know” suggestions after events, sharing user data with third parties, and finally customization of news feed. Customized advertisements are particularly useful in raising the users' awareness about the shared memories included in personalizing their social media experience as shared memories are most recognizable in this SIPM. In this sense, SIPMs can be considered a form of ambient memory augmentation systems as users try to recall the shared memories that triggered the SIPM. However, the relevance of the shared memories to the cued offline interactions remain mysterious to the majority of our user group. We also found that negative platform perception does not translate directly to an active reduction of the platform usage. We postulate this is attributed to the collectivist nature of the Egyptian culture that emphasizes the need to stay connected over other values. Our work informs the community about users' expectations about implicit usage of their shared memories in smart memory vaults. Additionally, looking at such patterns gives us a glimpse of future patterns for niche memory-altering technologies such as lifelogging.

2 METHODOLOGY

We investigated *the impact of surprising intelligent Facebook behaviors on the usage pattern and the affect towards the platform*. We use the term “surprising incidents” as a proxy for the strongest ambly remembered interactions involving traces of memories the user supplied to the social platform earlier but did not expect their utility in later interactions. To achieve this, we used a critical incident-recall qualitative questionnaire to reflect on a specific surprising incident when using Facebook, its consequences, perceived reasons and the associated affect with it.

2.1 Research Method: Online Survey

The survey was composed of three sections. Table 1 shows sample questions.

Demographics We inquired about nationality, length of Facebook usage and Computer Science expertise. The demographics helped us dissect and group the qualitative data.

¹Artificial Intelligence

Critical incident report We asked the participants to reflect on a specific surprising incident mentioning why it was surprising, its consequences and their interpretation of the reasons for it.

Attitude towards platform We used Self Assessment Manikin (SAM) to evaluate the affect of the participants about the incident [6]. Participants were asked if the incident changed their usage patterns towards Facebook, and if they considered the incident to be a privacy invasion.

We reduced the *self-report bias* by using neutral connotations about the platform and the incidents to avoid cuing the participants. Additionally, critical incident recall has been shown to reduce inaccuracies in self report as it focuses on a single incident rather than open ended questions left for a participant's emotion and interpretation at the time of answer [5]. Finally, we encouraged the participants to report more than one incident consecutively after reporting all details of the previous incident, to reflect holistically on the platform experience.

2.2 Procedures

After obtaining their consent, participants were asked if they were Facebook users. Only Facebook users were allowed to proceed with the questionnaire. They were then directed to the questionnaire that was divided into three pages corresponding to the parts without adding a title to the questionnaire page. By the end of the questionnaire, we asked the participants if they would like to report further incidents. Participants who agreed were redirected to the beginning of the questionnaire while others were directed to a page to voluntarily participate in a raffle for compensation.

2.3 Participants and Recruitment

We distributed the questionnaire through online channels such as popular Facebook groups to reach Egyptian users. We received a total of 57 responses, whereby four entries were excluded as the answers did not reflect on a specific incident ($N=53$). Participants were compensated with a raffle for a shopping voucher. There were 11 males and 42 females in the final corpus. Participants were aged between 20–56 years old (mean= 31.2, $SD=10.75$).

2.4 Analysis Methods

We used thematic analysis to analyze the qualitative data. Two researchers independently open-coded 20% of the entries using inductive coding techniques. In each incident, we were looking for a category for the surprising incident, its consequence on the user, and why the incident was surprising. Next, the tags were discussed and grouped into a single coding tree focusing on values coding to identify themes of incidents. Afterwards, one researcher tagged the full corpus using the defined tags and marked entries that required new tags. Each entry was assigned exclusively to a single tag. The new tags were discussed between the two researchers and the controversial entries were categorized collaboratively. Code instances were counted to define emerging themes (values coding). We did in-vivo coding and considered quotes from codes with at least two instances. We also did versus coding over the defined codes with respect to the demographic data we collected to identify

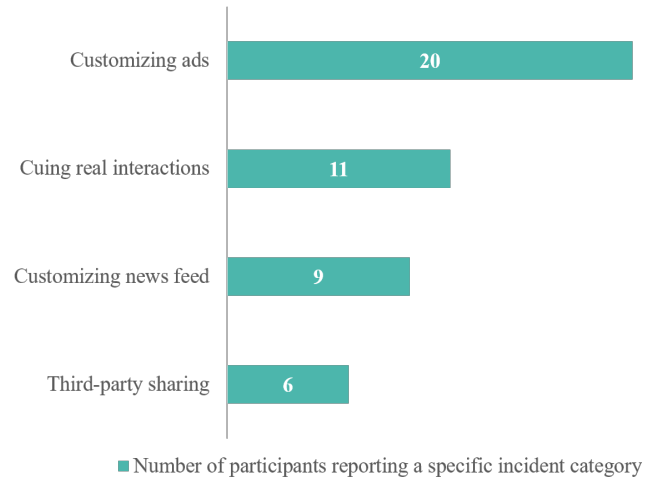


Figure 2: The emerging themes from the reported incidents sorted descendingly by incidental frequency.

differences across the user groups. We used medians to evaluate Likert-items data such as the magnitude of surprise and SAM scales.

3 RESULTS

We reflect in this section on three points: (1) categories of surprising incidents on FB, (2) impact of the surprise on the affective perception on FB, and (3) users' perception of the incident with respect to privacy violations.

3.1 Understanding Recalled SIPMs: What, Why, Impact?

We describe here the common types of SIPM that participants recalled and reflect on the top two reasons of the surprise, the consequences of the incident, and the associated affect and perception of the incident. Figure 2 summarizes the incidents' frequency. Table 1 summarizes the qualitative data about the top categories.

3.1.1 Personalized Customization of Advertisements. This category was reported the most in the survey data. Participants commented repeatedly on showing advertisements based on search activities in other platforms, conversations with friends on Facebook, or real verbal communication next to mobiles. For example, **P10** commented that *"It has personalised ads to what I have looked up recently, even on other platforms. Sometimes the ads are relevant to something I have talked about near my phone, but never even searched"*.

Participants were surprised mostly because they perceived it as exploitation for their data (7 participants). **P10** explains *"it seems like I'm being spied on, then manipulated by having suitable ads for my recent interests so I would spend money"*. Another theme was the lack of informed consent (6 participants), justifying why participants felt exploited. Such incidents resulted in a negative company image (10 participants), and increasing the awareness of the participants that their data was being used (3 participants). However, a minimal number of participants took an active action and reduced their platform usage (3 participants).

Table 1: Overview of the survey questions.

Type	Question
Open-ended	Please complete the following sentence and use this incident to answer the rest of the survey. Facebook uses our data or actions in different ways. I was surprised when I saw that Facebook ...
Likert-Item	On a scale of 1 (not surprised at all) to 5 (extremely surprised), how surprised were you by this incident?
SAM scales	What were your feelings after the incident?
Open-ended	Do you consider this incident to be a privacy invasion or not? Please answer with a yes or no then justify your answer.

3.1.2 Cuing Interactions with People. This category covered incidents of automatically suggesting new friends on the platform and limiting or extending their interactions with existing ones (11 participants). For example, **P7** commented that FB “*removed members from my makeup group by itself*”. Alternatively, **P20** commented that FB “*pops up people I may know instantly while I am just sitting beside them in the same place without even really knowing them*”.

The incidents were surprising because the participants did not understand the rationale behind them and how they were inferred from their platform interactions (7 participants). For example, **P50** comments on a new friend suggestion saying “*the person is nowhere near my circle or close to the people that I know and the number of mutual friends is very low*”. There was no prominent trend in the consequences of these incidents. Consequences included negative company image (4 participants), and starting conversations with others about the phenomenon (2 participants). However, the suggestions rarely resulted in creating a new virtual relationship (2 participants). We informally observed through our prior research that laymen are intrigued specifically by understanding how the suggestions are formed for new relationships and seem to relax and enjoy the potentially simple heuristics like geographical co-location. However, the results suggest that the recommendations do not translate to actions in the virtual world as expected.

3.1.3 Sharing Personal Data with Third-Party Entities. This category covered reports about utilization of platform data for the benefit of the third parties. It occurred repetitively (9 participants). However, the reported incidents were short and vague compared to the other categories. For example, **P2** said that the platform “*was selling data*”. Terms like “selling”, “using”, and “sharing” data were commonly used in the incident report.

Participants could not articulate clearly the reasons for finding those incidents surprising. Thus, there was no prominent trend and the reasons included inexplicable behavior from the platform, defying the platform objective, and exploitation. For example, **P19** said “*I was not aware that Facebook can monitor my actions*” while **P2** commented that “*it (selling data) is a big risk considering the fallout and the bad publicity*”. Similarly, there was no clear trend in the perceived consequences of the incidents. They varied between increasing the negative company image without behavioral change (4 participants) and reducing platform usage (4 participants). Overall, participants were not aware about data sharing policies, which is a challenge for informed consent.

3.1.4 Personalized Customization of News Feed. This category covers incidents about altering the amount of curated content from particular people and/or topics in the news feed (6 participants). The majority of reports were about topic customization rather than

people. For example, **P8** explains “*I was texting my friend on messenger and I kept seeing posts of the same topic on my News Feed*”. We consider this a form of ambient memory augmentation as participants realize the changes and missing/added topics in their news feed.

The primary reason for the surprise was not understanding the rationale behind the customization (3 participants), similar to the “cuing interactions with people” category. Terms like “bizarre” and “creepy” were frequently used to describe the interaction. Due to the limited number of reports in this category, other reasons like losing autonomy over displayed content and lack of informed consent occurred only once. There were no clear trends in the consequences of the incidents. Examples included increased awareness without reported behavioral change (2 participants) or active reduction of platform usage (2 participants).

3.2 Impact of Demographics on Recalled SIPMs

This section correlates the gender, age, and computer training with the categories of the reported incidents. We are cautious in speculating about those trends because of the unbalanced sample size. However, we report them to nudge further future investigations in this direction.

3.2.1 Gender. We observed that no males reported incidents related to cuing interactions with people, while 26% of the females did. Additionally, 45% of the males reported incidents related to sharing data with 3rd parties, while only 10% of the females reported about this category. This indicates that the females were more sensitive towards surprising social interactions while the males were more sensitive towards interactions resulting in monetary gain for the platform.

3.2.2 Age. The (40-60 years) segment mostly reported about advertisement customization. The frequency of reports shows the lack of awareness about personalization techniques in older generations despite recognizing the SIPM. However, the (30-50 years) segment focused their reports on sharing data with 3rd parties. This may be attributed to the younger generations being following news about public malicious use cases in contexts like politics.

3.2.3 Computer Science Training. The most prominent finding about groups with higher training is that they did not react aggressively to the SIPMs, as if they were expected. For example, **P49** when asked about the consequences stated “*Nothing, it’s fine for data analysis*”. This could imply the importance of technical literacy for educated guesses about the algorithms’ heuristics.

Table 2: Summary of the affect towards SIPMs on Facebook in Egypt. Participants were surprised about direct personalization of platform content compared to cuing offline interactions. They were particularly unsatisfied by advertisement manipulation and utilizing their data outside the platform. They also felt strongly about sharing their data with third parties and cuing their offline interactions such as proposing people they may know. () marks important patterns.**

Metric	Customizing ads	Third-party sharing	Customizing news feed	Cuing real interactions
Surprise	4 (surprised) **	4 (surprised) **	4 (surprised) **	3 (neutral)
Pleasure	3 (unsatisfied) **	2.5 (unsatisfied) **	2 (neutral)	2 (neutral)
Arousal	3 (neutral)	3 (neutral)	2.5 (neutral)	2 (dull) **
Dominance	3 (neutral)	2 (powerful) **	3 (neutral)	2 (powerful) **

3.3 Affective Perception of the Incidents

Table 2 summarizes the self-reported scales about how surprising the SIPMs were and the recalled affect after.

3.3.1 Surprise Magnitude. Participants were surprised equally (*Median* = 4) about customizing the advertisements, sharing data with third-parties and personalization of the News Feed. However, their reaction was neutral (*Median* = 3) towards cuing interactions with people.

3.3.2 SAM - Pleasure. This metric describes the level of pleasure (valence) associated with the reported incident. Customization of advertisements (*Median* = 3) and sharing data with third-parties (*Median* = 2.5) resulted in an unsatisfied reaction on the pleasure scale. On the other hand, customizing the news feed (*Median* = 2) and cuing interactions (*Median* = 2) led to a neutral reaction.

3.3.3 SAM - Arousal. This metric describes how interesting the incident was perceived. Participants were neutral about customization of advertisements (*Median* = 3), sharing data with third-parties (*Median* = 3), and customizing the news feed (*Median* = 2.5). However, they found cuing interactions with people (*Median* = 2) to be dull.

3.3.4 SAM-Dominance. This metric describes the prevalence of the reported emotions about the incident. Participants had powerful views on sharing data with third-parties (*Median* = 2) and cuing interactions with people (*Median* = 2), unlike the other two neutral categories (*Median* = 3).

3.4 Users' Perception of Privacy Violations

Forty seven participants mentioned that they perceived the SIPM as a privacy invasion while only six participants mentioned that they did not. Top reasons included unauthorized use of personal content in an unintended context. For example, **P50** stated that *"it felt like they know what I do and where I'm at always and acts upon it, an app should have no access to such information, it's there for me to use, not for it to use me"*. Others reported that some data was sensitive because they were unaware that "private" resources are being used. For example, commonly reported cases included the use of mics during conversations or Google search queries to access information and generate advertisements. This leads to cautious usage and sometimes behavioral changes in the usage patterns due to uninformed consent.

Thirty seven participants experienced a change in affect towards the platform while only twenty three reported a change in their behavior on the platform. This highlights the cycle of generalizing

beyond singular incidents to forming an emotion then actual behavioral change. This suggests that the number of users on the platform does not directly reflect user satisfaction. The most prominently reported changes were reduced feeling of privacy and reduced trust.

Participants who changed their usage behavior reported decreased sharing about specific topics or limiting sharing objectives to texting objectives. In addition, they also reported priming towards a specific topic or person. For example **P25** said *"I became more alert of the things that come up on my news feed now"*.

4 REFLECTIONS AND LESSONS LEARNT

4.1 Affect and Behavioral Change are Independent

Participants were more susceptible to changing their behavior when they perceived direct threat to their data like sharing it with 3rd parties, or if their curated content was directly affected like customizing their news feed. However, the low percentage changing their behavior despite the relatively high percentage reporting negative affective responses shows that the number of users on the platform does not directly reflect user satisfaction.

4.2 Sharing Data to Generate Revenue is Misperceived

Sharing data whether by providing it to third parties or through customized advertisements requires further research into effectively involving users, as this resulted in the strongest negative affective responses. We speculate that the users negatively reacted towards actions that were perceived to bring direct monetary value to the company without involving them. This could also be explained by linking it to Hsu et al.'s work [8] showing that users felt they could not control the privacy settings for customizing ads, even though they felt comfortable adjusting the settings for the news feed and notifications.

4.3 Identify Relevant Personalization Use Cases

The low arousal in all categories may imply low interest of users in detailed personalization compared to the emphasis of the technical community on enhancing its accuracy. In particular, cuing interactions with people was not positively perceived by the users despite being highly detectable by them.

5 CONCLUSION AND OUTLOOK

In this paper, we present a preliminary online survey (N=53) exploring what Egyptian users remember about their Smart Interactions involving Personal Memories (SIPMs) while using Facebook (FB). This work aids other researchers identify user expectations about implicit utilization of their shared memories on smart memory vaults to personalize their experience. We acknowledge the small sample size of our exploration. However, it is meant as a step forward to further include underrepresented non-WEIRD communities in privacy research of memory vaults. Our results suggest that users are still apprehensive about implicit utilization of their shared memories despite being part of the user-agreement to use free digital services. Thus, further research explorations are required to define sensitive data snippets particularly in the context of memory cues. The results also suggest that personalized content acts as a form of ambient memory augmentation solution. This refers to the users' ability to recall past interactions on the platform or shared content that trigger the personalization. Thus, they remember more about what they are doing. We also observed that cuing interaction with people through features like "people you may know"² could become a form of ambient memory implantation solution giving a fake sense of knowing the person after repetitively showing information about them. Additionally, we learnt that users were not excited about personalized interactions even though they were able to detect them. Our work is a step forward in highlighting challenges related to data governing to increase the social acceptability of smart memory vaults.

ACKNOWLEDGMENTS



This work is partially conducted within the Amplify project (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement no. 683008).

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²A feature that frequently proposes people that the platform thinks should be in your social circle by showing the name and photo of the person and the common friends.